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Research Article

Structural and Transport Properties of Norbornene-Functionalized Poly(vinyl alcohol) "Click" Hydrogel: A Molecular Dynamics Study

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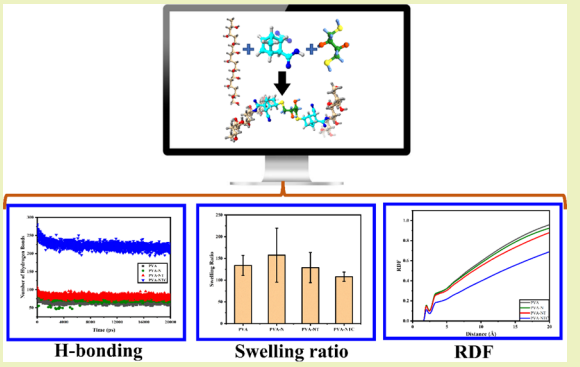
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ABSTRACT: The thiol−norbornene cross−linked poly(vinyl alcohol) (PVA) "click" hydrogel is a synthetic and sustainable polymer chain that is widely used in the biomedical field. Molecular dynamics simulations are invoked to measure the hydrogel's transport and structural properties that cannot be measured precisely by experimental techniques. There are three different forms of thiol−norbornene cross−linked PVA "click" hydrogel models constructed by replicating a representative unit cell of cross−linked PVA chains and immersing it into an aqueous environment. Additionally, the dynamics is compared with the movement of the PVA hydrogel in water. The AMBER14SB force field is utilized to describe the interatomic interactions. The system's dynamic behavior is achieved in the grand canonical ensemble and the isobaric isothermal ensemble. The dynamics of the system exposes the vital role played by norbornene in securing the stability of PVA polymer chains, thus contributing significantly to the overall robustness of the hydrogel. Delving into the structural analysis of the radial distribution function provides insights into the emergence of compact clusters, where interactions between PVA polymer chains and water drive their formation. The PVA hydrogel functionalizes with norbornene and exhibits a superior swelling ratio compared with the other variant hydrogels considered here owing to the freezing water phenomenon. Conversely, the swelling ratio of the cross−linked PVA hydrogel diminishes due to the intricate process of structural self−assembly. Consequently, the cohesive force within the cross−linked PVA hydrogel structure intensifies, highlighting the profound influence of bolstered hydrogen bonding. The exploration from this study provides valuable insights into the behavior of thiol−norbornene "click" PVA hydrogels and the importance of chemical cross−linking. The thiol−norbornene "click" PVA hydrogel holds tremendous potential for applications across diverse scientific domains, heralding a new era of possibilities.

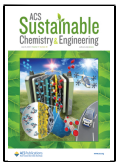


KEYWORDS: *molecular dynamics simulations, hydrogel, poly(vinyl alcohol), chemical cross−linking, norbornene click chemistry*

1. INTRODUCTION

The poly(vinyl alcohol) (PVA) hydrogel is an artificial polymer that has sparked a lot of interest in the biomedical field due to its benefits such as biocompatibility, nontoxicity, hydrophobicity, biodegradability, chemical stability, low cytotoxicity, and affordability.[1] The myriad hydroxyl moieties on the PVA polymer chains may lead to the creation of physically/chemically cross−linked PVA hydrogels that have applications in a variety of sectors.[2−4] The physical cross−link PVA gels are created by the freeze−thawing method, lack control over the degree of the cross−link, and have poor mechanical strength.[5] On the other hand, chemically cross−link PVA hydrogels facilitate a controllable cross−link network and higher mechanical strength.[6] The chemical cross−link is conveniently achieved by functionalizing the hydroxyl moiety with aminoxy,[7] hydrazide,[7] acrylamide,[8] thiol,[9] and semicarbazide,[7] cysteine moieties with maleimide[7] and 2−halo acetyl.[7] Apart from that, photo-initiated thiol−ene "click" reactions have recently gained attention for control network hydrogel formation.[2,10,11] The thiol−ene "click" hydrogels possess unique properties such as high water content,[12] excellent biocompatibility,[13] and tunable mechanical strength.[14] They have found applications in various fields, including tissue engineering,[15] drug delivery,[2] and biosensing.[16] The thiol−ene cross−linked PVA hydrogels offer improved stability and long−term performance compared to other hydrogel systems.[2] The ability to control the gelation process and tailor the hydrogel properties has opened up new avenues for research and development. Their versatile nature and promising characteristics have sparked considerable interest and exploration, making them a valuable asset in advancing scientific understanding and technological innovations.[17,18] PVA is thus being functionalized with norbornene

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(an "ene" moiety) to produce a "click" hydrogel by thiol−ene "click" chemistry.[9,18,19]

The primary objective of the norbornene−functionalized PVA reacting with thiol is to create a robust and stable cross−link network with high reactivity, stability, cytocompatibility, and promising candidates for biomedical applications. The stability of the macromeric thiol−ene formulations has been analyzed and is found to be sufficient when stored with 100 ppm of pyrogallol for up to 30 days.[9] The visible light−polymerized thiol−norbornene hydrogels are used for pancreatic ductal adenocarcinoma cell research. Researchers observed that it is an efficient and cytocompatible way to in situ cell encapsulation. These modular biomimetic hydrogels allow for facile tuning of stiffness and are susceptible to enzyme−mediated degradation.[19] The structural moieties, morphology, rheology, thermal, and mechanical properties of the hydrogel are characterized by experimental techniques.[20−23] However, the hydrophilic nature of the network resists the precise measurement of such physicochemical properties and often lacks reproducibility. Further, the experimental techniques cannot enlighten the structural orientations in solution medium, dominant nonbonded interaction such as hydrogen bonding, and transport properties of the scaffold. Hydrogen bonding plays a prominent role in the application of hydrogels, which can be accurately mapped through theoretical investigation. In contrast to experimental simulations, the molecular dynamics (MD) approach can replicate gel microstructures and mimic material behaviors at the molecular level, providing a wealth of information about the microstructures. MD simulations adequately reveal such properties.[24−26] Following this direction, MD simulation of PVA hydrogel has revealed hydrogen bonding (H−bonding),[27,28] diffusivity,[27] conductivity,[27] and stress−strain[28,29] behavior.

The dependency of mechanical properties such as tensile strength on temperature and absorbed water molecules are adequately described by a physically cross−linked PVA gel through MD simulations.[28] Additionally, the H−bonding network is mapped in that investigation. The relationship of H−bond evolution and stability with temperature can be adequately mapped through molecular trajectories.[24−26] Intramolecular structural distribution leads to the aggregation behavior of the PVA gel network. The temperature dependency of such behavior is reported in the literature.[30] The MD simulation investigates the composition of nonfreezing water, free water, and freezing water within an irradiation cross−linked PVA hydrogel. This study explores the dependencies between these water components and the cross−linking process of the hydrogel.[31] The understanding of hydrogelation processes, mechanical characteristics, and the encapsulation/adsorption of various materials has been aided through MD research.[32] The interaction between the liquid medium and the chemically cross−linked PVA gel and the topological constraint at the cross−linking junction are explored by MD simulations.[33] Looking at the literature, physically cross−linked PVA gels have received more attention than tailorable chemically cross−linked hydrogels. This may be attributed to the structural complexity that a chemical cross−linker poses on the various moieties.

Being a tailorable chemically cross−linking structure, the network poses a challenge to the all-atom MD simulations. This is the exact objective of this study, i.e., exploration of structural and transport behavior of a chemically cross−linked norbornene−functionalized PVA hydrogel. As per our knowledge, this is the first attempt in the theoretical investigation of the thiol−ene "click" PVA hydrogel. The atomistic model of the thiol−ene-functionalized PVA gel with appropriate water content will be constructed. The structural stability is observed by the root-mean-square deviation method. The evolution of the hydrogen bond network and dominant interactions between water and the polymer network in the gel is being explored. The distribution of functional moieties of the polymer backbone and the mobility of functional moieties are being analyzed. The mobility of the hydroxyl of the polymer backbone is also investigated.

2. COMPUTATIONAL METHODOLOGY

The atomic structure of the vinyl alcohol monomer is drawn using the Avogadro package.[34] The same package is further used to create a PVA oligomeric unit with 10 vinyl alcohol monomer units. Thereafter, to avoid steric hindrance, the fifth hydroxyl moiety is functionalized with norbornene. The thiol moieties are included in the "ene" functionality of norbornene to create thiol−ene-functionalized PVA. The representative structures of these three are given in Figure 1. Later, PVA, norbornene−functionalized PVA (PVA−N, Figure 1b), and dithiothreitol−norbornene−functionalized PVA (PVA−NT, Figure 1c) replicated eight times by in-house programs (Figure S1). Therefore, each of the chains contains 80 repeat units of the vinyl alcohol monomer with norbornene, and dithiothreitol−norbornene functionalization is at every 10th repeating unit. However, dithiothreitol acts as a cross−linker, and from

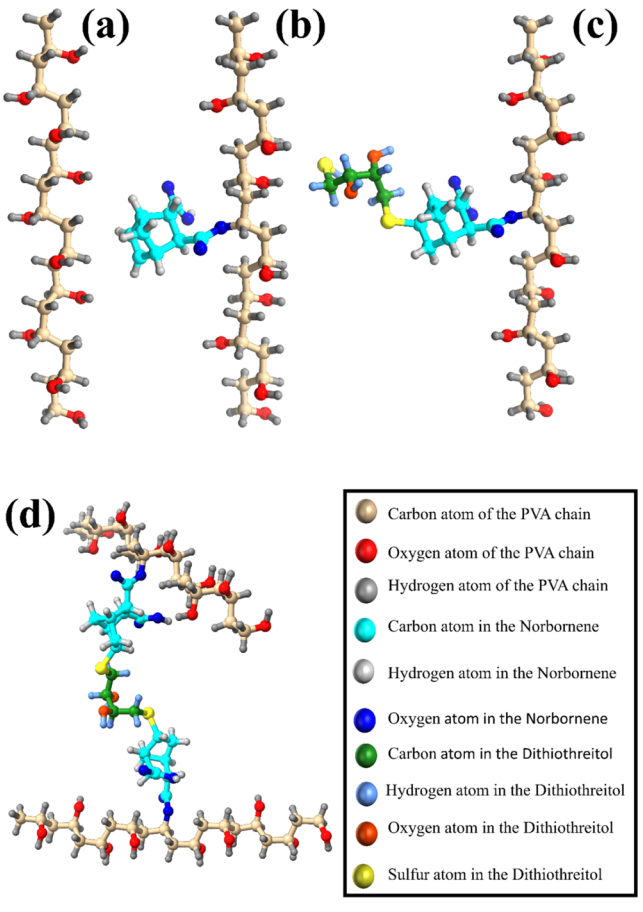


Figure 1. Initial structure of polymers: (a) PVA, (b) PVA−N, (c) PVA−NT, and (d) PVA−NTC.

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the experimental evidence, it is established that dithiothreitol connects the norbornene−functionalized chains to form a gel network.[9] Hence, the alternative chains of PVA−N and PVA−NT are joined to form a thiol−ene cross−link PVA gel (PVA−NTC, Figure 1d). A couple of such cross−linked gels are connected to form an enclosed network shown in Figure S2.

The objective of the work is to study the dynamics of PVA and its thiol−ene-functionalized polymers. Hence, four different sets of configurations, i.e., PVA, PVA−N, PVA−NT, and PVA−NTC, are immersed into the water using the Packmol package.[35] The TIP4P water model has been implemented for the water molecules.[36] The initial number of molecules in each system and the corresponding dimensions of the box are given in Table S1 as the Supporting Information. The partial charges on each atom are generated by the Gasteiger method.[37] The Amber14SB force field[38] is used to correlate the bonded and nonbonded interactions. The Amber14SB force field is an updated version of the AMBER99SB force field, which has wider applications in macromolecule simulation.[39−42] Time-resolved trajectory through MD simulations has been performed in the GROMACS-2022.3 package.[43] The functional form of the Amber14SB force field is given by eq 1.

$$V_{\text{total}} = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} k_\chi[1 + \cos(n\chi - \delta_\chi)] + \sum_{\text{improper}} k_\Psi[1 + \cos(n\Psi - \delta_\Psi)] + \sum_{i=1}^{N-1}\sum_{i<j}^{N}\left\{4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}\right\} \quad (1)$$

where V_{total} represents the total potential energy which is represented as the sum of bond stretching energy, angle bending energy, torsional energy for proper dihedral and improper dihedrals, van der Waal interaction energy, and Coulombic interaction energy. K_b is the bond stretching constant, r is the length of the bond and r_0 is the standard length of the bond, k_θ is the bending angle energy, θ is the angle between the molecule, θ_0 is the standard angle, k_χ is the proper dihedral constant, χ is the dihedral angle between two planes, n is the multiplicity (the number of equivalent minima in the energy surface), δ_χ is the phase shift, k_ψ is the improper dihedral constant, ψ is the improper dihedral angle between the planes, n is the multiplicity factor, δ_ψ is the phase shift, ε_{ij} is the depth of the potential wall, σ_{ij} is the distance at which the potential is zero, r_{ij}^{-12} is the repulsive term which dominates at a short distance, r_{ij}^{-6} is an attractive term which dominates at a long distance, ε_0 is the permittivity of vacuum, q_i and q_j are the charges of two particles, and r is the distance between the two charge particles.

The dynamics of the systems is obtained through the successive methodology of energy minimization, equilibration, and production. The steepest descent minimization algorithm is utilized for energy minimization, with an energy threshold for a cutoff of 10 kJ mol^{-1} N m^{-1}. This ensures bad contact between various stages of MD simulations. A maximum number of 500,000 steps with a step size of 0.001 is provided. Periodic boundary conditions are considered for all the three dimensions. The cutoff distances of 1.0 and 1.2 nm are used to compute the van der Waals dispersion interactions and a long-range electrostatic interaction, respectively. Long-range electrostatic interactions are computed using the particle mesh Ewald technique.[44] The integration of the motion equation is performed using the Leapfrog algorithm[45] with a 2 fs time step. For equilibration and production run, the velocity rescale[46] approach is used to couple the temperature. This method has been thoroughly examined and is well acknowledged in the literature for the simulations of liquid systems.[47−49] Equilibration is performed in the *NVT* ensemble for 5 ns, followed by a 5 ns *NPT* run at 310 K. Using the Parrinello−Rahman method,[50] the pressure is linked to 1.013 bar during the *NPT* equilibration phase. The velocity-rescale algorithm is used to regulate the temperature during *NPT* equilibration. The LINCS algorithm's[51] constraints are applied to freeze the vibration of hydrogen-containing moieties in the equilibration phase. The *NPT* ensemble has been considered for the production run of 20 ns at 310 K and 1.013 bar using the Parrinello−Rahman barostat and the velocity−rescale thermostat. Simulation trajectories are saved at 10 ps intervals. This is determined to be adequate to produce precise findings of the statistical averaging of a single trajectory for the polymer chains in the liquid state. VMD and Chimera X packages[52−54] have been used for the visualization of trajectory, and analysis is carried out using in-house Python scripts. The different time interval snapshots of the simulation are displayed in Figures S3 and S6 of the Supporting Information. The snapshots of the systems at the end of the simulation are shown in Figure 2.

2.1. Analysis of Trajectory. *2.1.1. Root Mean Square Deviation.* Root mean square deviation (RMSD) is a common measure used in MD simulations to quantify the difference between the two structures. In MD simulations, atoms are allowed to move and interact with each other based on the physical laws, which can lead to structural changes over time.

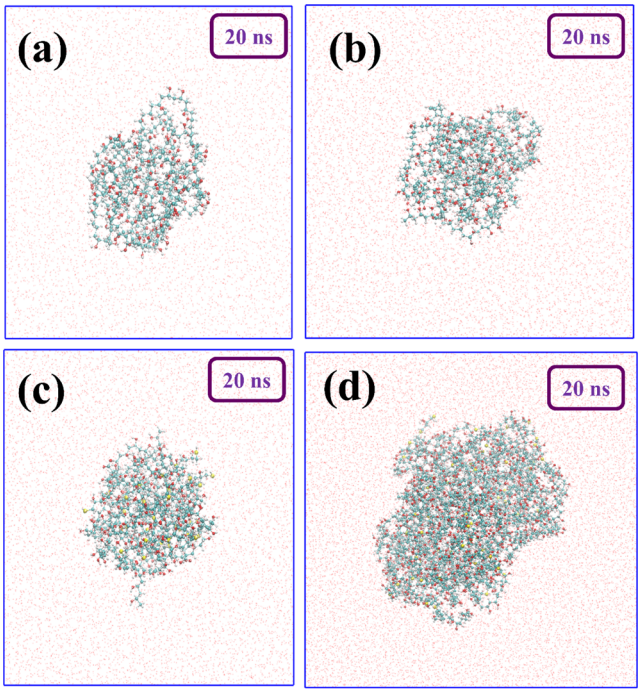


Figure 2. Final structure of hydrogels at the end of 20 ns: (a) PVA gel, (b) PVA−N gel, (c) PVA−NT hydrogel, and (d) PVA−NTC hydrogel.

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RMSD is calculated by aligning two structures based on a chosen reference structure and then measuring the root-mean square of the distance between the corresponding atoms in the two structures. It represents the average distance between the atoms in the two structures, taking into account both the magnitude and direction of the displacement. The common formula used to determine the RMSD of specific atoms in a molecule from a reference structure is

$$\mathrm{rmsd}(t_1, t_2) = \left[\frac{1}{M}\sum_{i=1}^{N} m_i \left\|r_i(t_1) - r_i(t_2)\right\|^2\right]^{\frac{1}{2}} \tag{2}$$

$$M = \sum_{i=1}^{N} m_i \tag{3}$$

where the ith particle's mass is given by m_i, N is the total number of particles present in the system, and r_i is the position at time t_1 and time t_2. RMSD is a useful measure for assessing the stability and convergence of MD simulations, as well as for comparing the structures obtained from different simulations or experimental data. It can also be used to identify structural changes and analyze the behavior of proteins, nucleic acids, and other biomolecules.

2.1.2. Radial Distribution Function. The radial distribution function (RDF) is a commonly used analysis tool in MD simulations that describes the spatial distribution of particles in a system. The RDF offers insights into the system's structure and interactions by quantifying the probability of locating a particle at a specific distance from another particle. The mathematical formulation for the RDF is as follows

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 \mathrm{d}r}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta(|r_i - r_j| - r) \tag{4}$$

where $g(r)$ is the RDF and N is the total number of particles in the system. V is the total volume of the simulations box, r is the distance between the reference particle and the other particle in the system, dr is the width of the radial bin, $\delta(z)$ is the Dirac delta function, which equals 1 if $z = 0$ otherwise, and $|r_i - r_j|$ is the distance between particles i and j. The RDF calculation involves dividing the simulation box into a series of concentric spherical shells with widths of dr. The particles in the system are then counted within each shell, and the result is normalized by the density of the system to obtain $g(r)$. The RDF can provide valuable insights into the structure and interactions of a system. The coordination number is a measure of the number of neighbor atoms or molecules surrounding the central atom or molecule. It is a key parameter in determining the local environment and properties of a system. It is calculated as

$$\mathrm{CN}(r) = 4\pi\rho N\int_0^r g(r)r^2\mathrm{d}r \tag{5}$$

where $CN(r)$ is the coordination number, ρ is the number of density particles, N is the total number of particles in the system, and $g(r)$ is the RDF. The integral over the spherical shell of radius r counts the number of particles within a distance of the reference particle.

2.1.3. Hydrogen Bonding. In hydrogels, hydrogen bonds play a key role in determining the properties of the polymer network such as its strength, elasticity, and swelling behavior. The strength of hydrogen bonds in hydrogels can be influenced by several factors, including the nature of the polymer, the degree of cross−linking, and the presence of functional moieties that can participate in hydrogen bonding. The hydrogen bond between an acceptor atom, A, and a donor atom, D, is done by using a geometrical criterion. When the hydrogen donor to acceptor angle (α_{HDA}) is less than 30°, and the distance between the donor and acceptor atoms (r_{DA}) is less than 0.35 nm, a hydrogen bond is said to have formed. The calculation of the dynamics of hydrogen bonding made use of the definition of the intermittent hydrogen bond autocorrelation function as

$$C_{\mathrm{HB}} = \frac{\langle H(\tau)H(\tau + t)\rangle}{\langle H(\tau)H(\tau)\rangle} \tag{6}$$

where C_{HB} is the number of hydrogen bonds and $H(t)$ is assumed to be either 1 or 0 depending on whether the chosen pair forms a hydrogen bond at that specific moment.

2.1.4. Swelling Ratio. The swelling ratio is a crucial characteristic of hydrogels because it governs the release patterns of solvents and drugs from these polymeric networks. Water absorption leads to a proportional increase in the hydrogel's weight and hence an increase in swelling ratio. A polymer can absorb water through two mechanisms: the incorporation of freezing water and the uptake of nonfreezing water. The first method is to define nonfreezing water as water molecules that are directly linked to the polymer chain via hydrogen bonds. Freezing water is a relatively weak interaction between water and polar functional moieties. It is defined as the number of water molecules that is directly connected to the nonfreezing water via hydrogen bonds. We used both kinds of water to calculate the swelling ratio. The swelling property can be computed as

$$\mathrm{SR}(\%) = \frac{W_{\mathrm{eq}} - W_i}{W_i} \tag{7}$$

where W_{eq} is the weight of hydrogel composites after swelling in water and W_i is the weight of dried hydrogel composites.

2.1.5. Mean Square Displacement. Self-diffusion is an extremely important characteristic of a hydrogel for microgel development. It is obtained by plotting the slope of the mean square displacement (MSD) of the molecules against time over a sufficiently long time period. The MSD describes the displacement of a particle or a molecule from its initial position. It is calculated by taking the difference between the position of the particle at a given time and its initial position, squaring that value, and then taking the average over all particles in the system. The following equation can be used to determine the MSD.

$$\mathrm{MSD}(t) = \langle(r_i(t) - r_i(0))^2\rangle = \frac{1}{N}\sum_{i=1}^{N}[(r_i(t) - r_i(0))^2] \tag{8}$$

where $r_i(t)$ represents the positions of atom ith at time t, $r_i(0)$ is its initial position, and the angled brackets ($\langle\ \rangle$) denote an ensemble average over all particles in the system. Using the Einstein relationship, the slope of the MSD can be used to compute the diffusion coefficient D.

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$$D = \frac{1}{6}\lim_{t\to\infty}\frac{\mathrm{d}(\mathrm{MSD}(t))}{\mathrm{d}t} = \frac{1}{6N}\lim_{t\to\infty}\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{N}[((r_i(t) - r_i(0))^2] \quad (9)$$

3. RESULTS AND DISCUSSION

3.1. Root Mean Square Deviation. Figure 3 represents the RMSD plot of PVA, PVA−N, PVA−NT, and PVA−NTC.

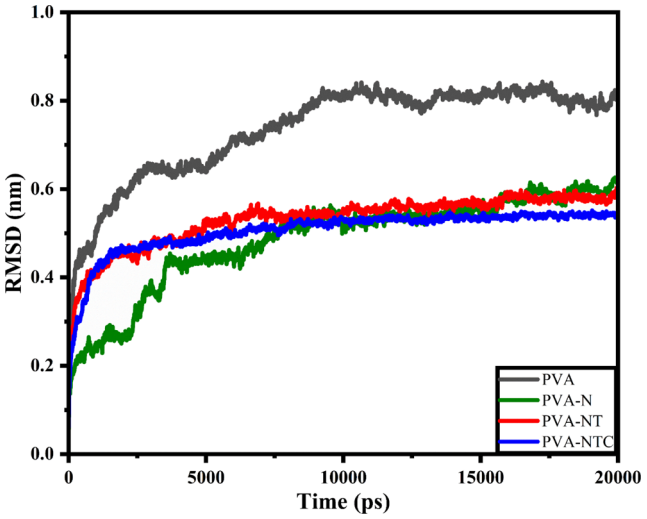


Figure 3. Polymer chain RMSD to its relative backbone.

For all polymers, it is evident that the RMSD value gradually increases with time until they reach an equilibrium plateau, after which they show a minimal fluctuation around their average value until the end of the simulations. Among the four RMSD, the PVA chain has a maximum RMSD value of 0.808 nm after 20 ns in solution. The final RMSD value further indicates the stability of the polymeric chains in water. Being the largest RMSD, PVA is the least stable among the four, and PVA−NTC shows the most stable behavior. The RMSD of PVA chains increases up to 10 ns, followed by minimal fluctuation in the solution for the rest of the period. After the functionalization of the PVA chain with norbornene, the RMSD value reduces to 0.52 nm after 20 ns. This behavior is attributed to the stabilization of PVA−N in solution. However, a gradual increment of fluctuation has been observed until 8 ns, followed by minimal fluctuation during the end of the simulation. Norbornene is showing a stabilization effect on the polymer chains due to its rigid structure, which can prevent the polymer chains from collapsing or under conformational change during the simulation run. The RMSD value of the PVA−NT polymer chain at 20 ns is 0.48 nm, which is lower than PVA and PVA−N chains. However, the gradual increment of fluctuation is only up to 5 ns. This result is also significant as it suggests that the addition of dithiothreitol in the PVA−N polymer chain can help to stabilize the system, and it requires less equilibration time than the other two polymeric structures. The RMSD of the PVA−NTC polymer chain takes 3 ns to reach the fluctuation plateau. After that, the hydrogel structure shows minimal fluctuation during the end of the simulations. The order of the attaining RMSD plateau is PVA−NTC < PVA−NT < PVA−N < PVA. This behavior confirms that the chemical cross−linking enhances the stability of the network more than the precursors. Therefore, the structurally integrated structure is better suited for encapsulation applications.

3.2. Hydrogen Bond. Hydrogen bonds (H−bonds) play a significant role in the structure and properties of polymeric materials. The strength of hydrogen bonds between polymer to polymer increases, which can lead to high mechanical strength, less swelling behavior, and fewer elasticity properties, which lead to forming a rigid hydrogel network. In the aqueous medium, the hydrogel can form H−bonds, which it is categorized as interchain (HB_{inter}), intrachain (HB_{intra}), and between polymer−water (HB_{P-S}). HB_{inter} is further classified depending on the presence of adjacent repeat units (HB_{adj}). Figure 4a shows the hydrogen bonds of HB_{intra} and HB_{inter}

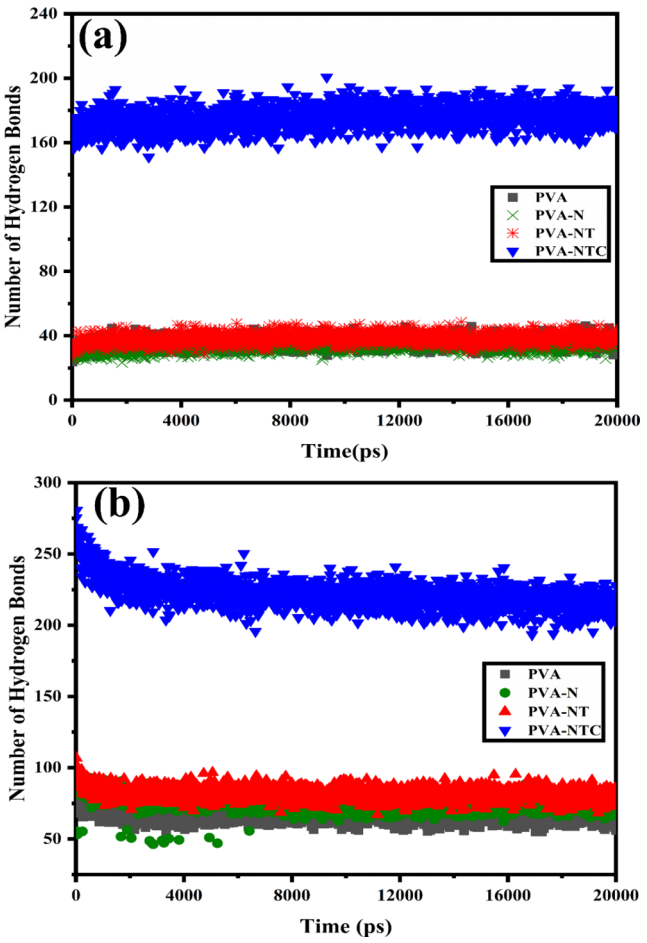


Figure 4. Number of hydrogen bonds in the system: (a) number of hydrogen bonds between polymer to polymer and (b) number of hydrogen bonds between polymer to solvents.

types. The hydroxyl moieties, norbornene, and dithiothreitol functional moieties act as H−bond donors and acceptors. The H−bond formation for all the four types of polymer chains increases initially and attains a constant after a certain period. The order of the H−bonds between the polymer chain to the polymer chain is PVA−N < PVA < PVA−NT < PVA−NTC. The average number of H−bonds between the polymer chain to the polymer chain is 34, 36, 38, and 175, respectively, given in Table 1. The PVA−N chain forms a lower number of H−bonds between the polymer chains, which can lead to high swelling properties and less mechanical strength. The PVA−

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Table 1. Average Number of Hydrogen Bonds

polymer name	average no. of hydrogen bonds between polymer	average no. of hydrogen bonds between solvents	percent of hydrogen bonds forming in polymer chain to the polymer chain	percent of hydrogen bonding forming in polymer chain to solvents
PVA	36	66	35	65
PVA−N	34	76	30.9	69.1
PVA−NT	38	82	32	68
PVA−NTC	175	222	44	55.9

NT polymer chains show on average 38 H−bonds per frame. The average number of hydrogen bonds between the PVA−NT chains increases significantly because of functionalization with dithiothreitol, which has a couple of hydroxyl moieties and one thiol moiety. However, the number of H−bonds is increased 4.5-fold for the chemically cross−linked PVA−NTC. The cross−linking brought the donors and acceptors in the polymeric chain closer, leading to the dense formation network of H−bonds.

Figure 4b represents the evolution of the H−bond among the donor and acceptor moieties of polymer chains and water. The initial behavior of H−bond formation between polymer−water contradicts that of polymer−polymer. The initial reduction of the H−bond for four types of polymer chains indicates that polymers tend to hold their structural integrity during immersion. However, after a certain period, it forms an H−bond with water. The order of H−bonding between polymer and water is PVA < PVA−N < PVA−NT < PVA−NTC. However, the number of H−bond with water has a higher proportion of 55−69% than that within the polymer network (Table 1). This refers to the penetration of water molecules into the gel network. The carboxylic (from norbornene) and hydroxide moieties are introduced to enhance the interaction with water, which leads to a significant increase in H−bonds. Dithiothreitol functionalization adds another donor moiety, leading to the further increase in H−bond. The average number of hydrogen bonds between the polymer chain to the water is 66, 76, 82, and 222, respectively, given in Table 1. The PVA−NTC polymer chain self-assembles by forming a H−bond with the surrounding water molecules, and the molecules penetrate into the gel network to attain stability.

3.3. Swelling Ratio. Swelling is a fundamental physical property of a hydrogel that determines its ability to absorb water. The swelling ratio can be used to determine other properties, such as mechanical strength and permeability.[55,56] Figure 5 depicts the swelling property of a hydrogel. The order of the swelling ratio is PVA−N > PVA > PVA−NT > PVA−NTC. The PVA−NTC polymer chain shows the least swelling behavior of 108.02 with a standard deviation of 10.7. This is because the PVA−NTC polymer backbone has a dense cross−link network. Due to the chemical cross−linking, the chain mobility decreases, and thus, the swelling also decreases significantly. It is observed that the PVA−N polymer chain has a high swelling ratio of 157.58 with a standard deviation of 62.26. This is because the PVA−N polymer chain is functionalized with *cis*-5-norbornene-*endo*-2,3-dicarboxylic acid, which has two carboxylic acid moieties and one unsaturated alkene. Owing to their presence, the PVA−N polymer chain can attract more freezing and nonfreezing water, which leads to a larger swelling ratio than the other three.

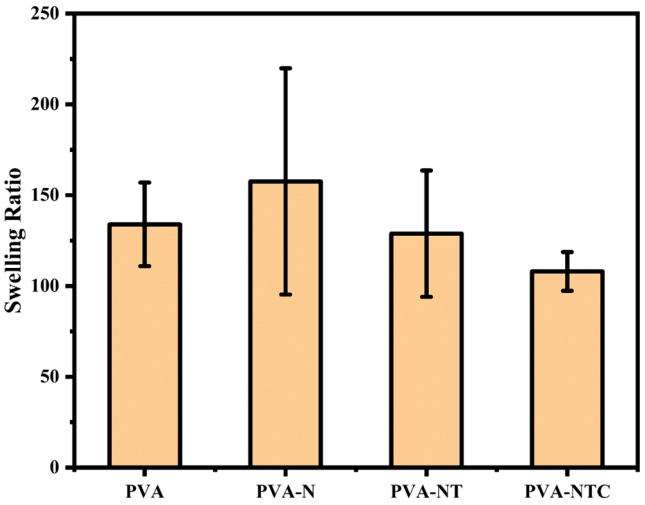


Figure 5. Swelling ratio of hydrogels.

While high swelling ratios can be advantageous in certain applications, such as in drug delivery or wound healing, they can also have certain disadvantages in terms of weakened mechanical strength. Additionally, the high swelling can decrease the rate of diffusion of solute through the hydrogel. The stability of the hydrogel is also affected by the high swelling ratio. The PVA−N polymer chain has the highest mobility among the other hydrogel models. The high chain mobility and the high swelling ratio can lead to the loss of structural integrity and mechanical stability of PVA−N hydrogel. The PVA−NT chain can form more hydrogen bonds with solution because of dithiothreitol, which has a couple of additional hydroxide moieties and hence can attract more nonfreezing water than freezing water.

3.4. Analysis of RDF. RDFs analyze the water distribution within the polymer network and atom distribution in polymer chains. In a homogeneous system, RDF values tend to converge to unity at large separations. However, the presence of solute clusters can disrupt this characteristic. Figure 6 shows the RDF curves ($g_{O-HW}(r)$) between the oxygen atoms present in the polymer backbone and the hydrogen atoms of the water molecule. Experimental evidence suggests that PVA has a tendency to form aggregate microdomains in hydrogels. It is worth noting that the $g_{O-HW}(r)$ value remains below unity, even at significant distances. This observation indicates the formation of entanglements and small-scale clusters in the system.

The results indicate that the presence of hydrogen bonds between the PVA polymer and the water molecules is crucial. These bonds play a vital role in determining the spatial distribution of the particles in the system. The first maxima in the RDF occur at a distance of 2 Å, which is in the hydrogen bond regime. The RDF values are 0.148, 0.156, 0.147, and 0.112 for PVA, PVA−N, PVA−NT, and PVA−NTC, respectively. The coordination numbers are 0.136, 0.145, 0.133, and 0.098, respectively. The RDF values for all of the systems are less than unity at large distances, indicating the formation of small clusters of chains. The maximum value of the RDF for PVA−N suggests that this polymer forms more hydrogen bonds with water molecules than do the other polymers, resulting in a higher probability of finding particles at a certain distance. This is also reflected in the coordination numbers, which are higher for PVA−N than for the other

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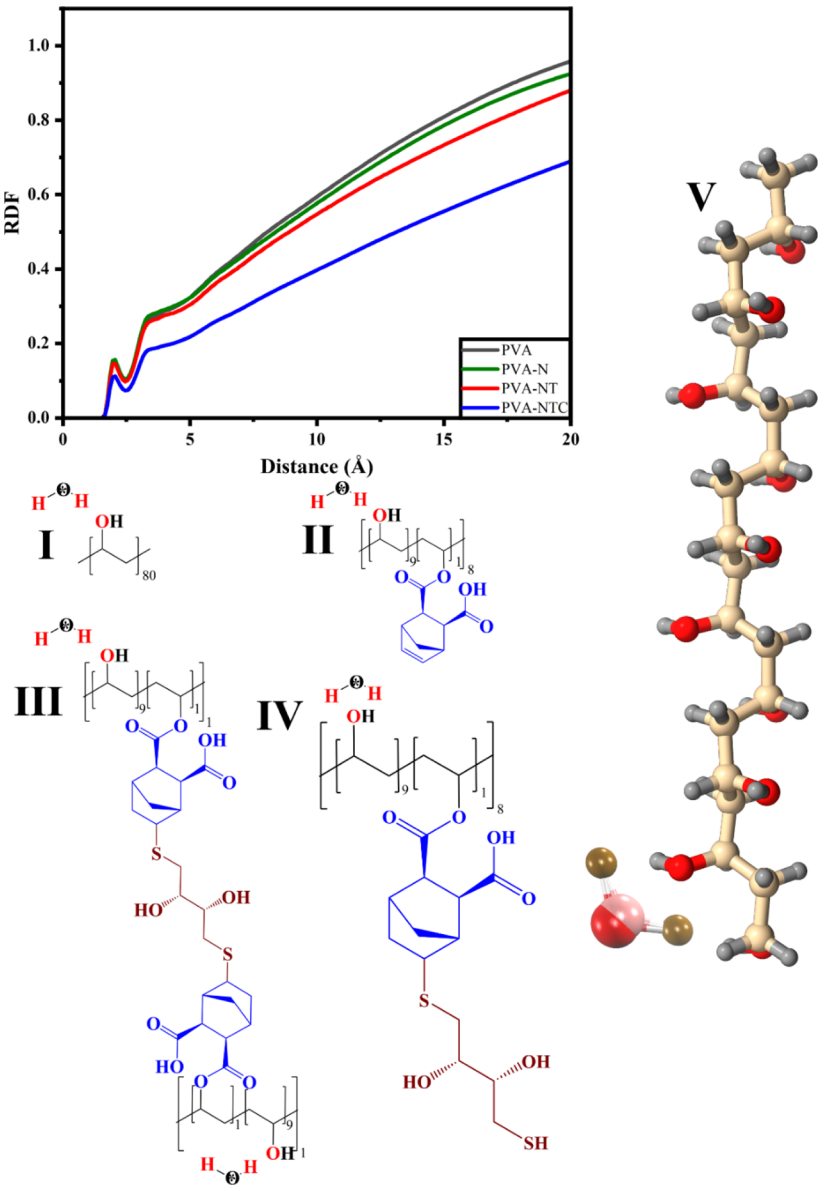


Figure 6. RDF of oxygen atoms in the polymer backbone to water hydrogen atoms. (I) Schematic representation of PVA interacting with water, (II) schematic representation of PVA−N interacting with water, (III) schematic representation of PVA−NTC interacting with water, (IV) schematic representation of PVA−NT interacting with water, and (V) 3D representation of PVA polymer chain interacting with water hydrogen atoms.

polymers. However, the increasing trend of RDF at a larger distance suggests the presence of hydrogen atoms of water in a region where the density is still lower than the average bulk density of the system. This could be due to the presence of spatial correlations between the particles, which result in a lower density at certain distances. Overall, the results suggest that the formation of hydrogen bonds between the PVA polymer and water molecules plays a crucial role in determining the spatial distribution of particles in the system and that the behavior of the RDF reflects the interparticle interactions and the structural properties of the particles.

Figure 7 represents the RDF curve between the conjugate double bond of norbornene, which is functionalized with the PVA molecule, and the oxygen atoms of the water molecule. The first maxima are observed at 3.95 Å, which indicate the interaction between the conjugate double bond in the functionalized polymer chain to the oxygen atoms of water. The RDF values are 0.390 and 0.201 for the PVA−N chain and PVA−NTC chain, respectively. The coordination numbers of these chains are 1.43 and 0.70, respectively. The PVA−N chain has a 2-fold higher peak than PVA−NTC, which suggests that the number of water molecules present near the PVA−N chain is higher than that present near the PVA−NTC chain. The coordination number also supports this inference. The PVA−NTC polymer chain also has unreacted thiol moieties, which can potentially disrupt the interaction between the polymer chain and water. Due to this reason, the PVA−NTC chain has a lower radial distribution value and a less sharp peak than the PVA−N chain. After that, both chains show a linear increase in RDF with distance. It signifies the finding that the water molecule is still lower than the average density of the system.

Figure 8 shows the RDF graph between the unreacted thiol moiety and the ene moieties of norbornene in the PVA−NTC polymer chain. There are three solvation shells observed in the

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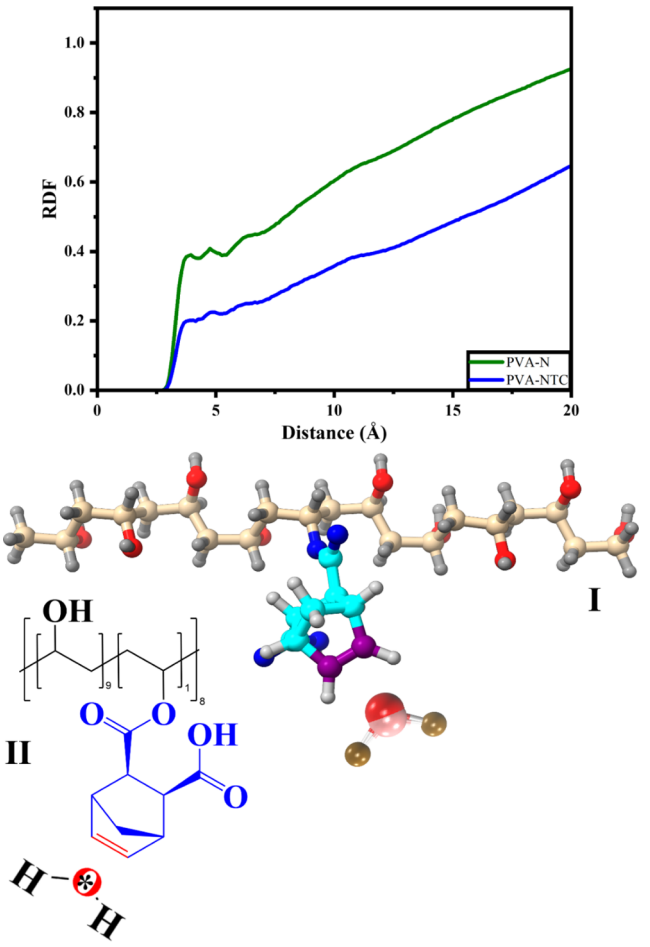


Figure 7. RDF of conjugate double bonds of norbornene and oxygen atoms of water. (I) 3D representation of the PVA−N chain interacting with oxygen atoms in the water. (II) Schematic representations of the PVA−N chain interacting with oxygen atoms of the water molecule.

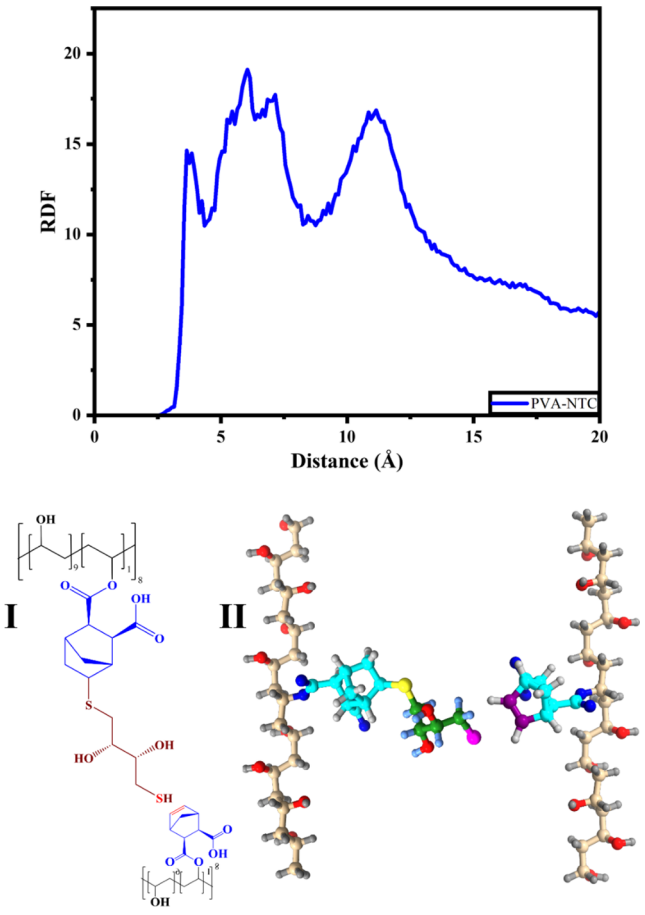


Figure 8. RDF between conjugate double bonds and unreacted thiols. (I) Schematic diagram of the PVA−NT chain interacting with the PVA−N chain. (II) 3D representation of the PVA−NT chain interacting with the PVA−N chain.

distribution. The first maxima are observed at 3.75 Å with an RDF value of 13.943 and a coordination number of 0.014. The first peak in the RDF curve indicates that there is a high probability of finding particles at a distance of 3.75 Å from each other. This peak may correspond to the distance at which van der Waals forces or other short-range interactions between the particles become significant. After the first peak, the RDF value decreases. The maxima for the second solvation shell are observed at 6.05 Å, with an RDF value of 19.116 and a coordination number of 0.192. The third peak is observed at 11.15 Å, where the RDF value is 16.871, and the coordination number is 1.327. This peak could be due to the Coulombic interaction or electrostatic interaction between those molecules. After the third peak, the RDF curve value decreases.

3.5. MSD and Diffusion Coefficient. Figure 9 represents the MSD plot of oxygen atoms in the polymer backbone (marked in red in Figure 9) of PVA, PVA−N, PVA−NT, and PVA−NTC polymer chains. The PVA polymer shows the second-highest MSD. It signifies the greater mobility of oxygen atoms in the polymer backbone. This reconfirms the larger mobility of PVA chains obtained from the swelling ratio and hydrogen bond analysis. There are five distinct regions visible in Figure 9a. In the first region, the MSD curve becomes concave downward with time for 7.0 ns, which signifies the subdiffusive behavior. The particles move slower than time. This phenomenon is observed because of the entanglement of the polymer chains with each other. These entanglements hinder the free movement of individual polymer chains, leading to subdiffusion. After 7 ns, the MSD curve shows a linear behavior up to 10 ns. This is primarily determined by the velocity gradient and the interactions with the surrounding solvent molecules, resulting in random movement of the chain. Such random motion leads to diffusive behavior of the chains. Here, the self-diffusivity of the oxygen atom is 1.68×10^{-10} m^2/s. After 10 ns, the particles start to move slower. As a result, the MSD curve becomes concave downward until 14 ns, which signifies the subdiffusive behavior. This is because of the interaction between the PVA chain to the PVA chain and PVA chain solvent molecules. After 14 ns, the particles start to move faster. As a result, the MSD curve shows a linear behavior up to 17.5 ns. During this time period, the logarithmic MSD slope is greater than 1. It signifies a superdiffusive behavior. This is because the polymer chains disentangle with each other and move freely. After 17.5 ns, the MSD curve shows more nonlinearity until the end of the simulation. This is because polymer movement is free and less collides with other particles.

Figure 9b,c shows the time-resolved linear increment of the MSD curve of the oxygen atoms in the PVA−N and PVA−NT polymer chains, respectively. The PVA−N polymer chain reveals the highest MSD curve. The PVA−N polymer chain shows four regions in the MSD curve. The first region is at the

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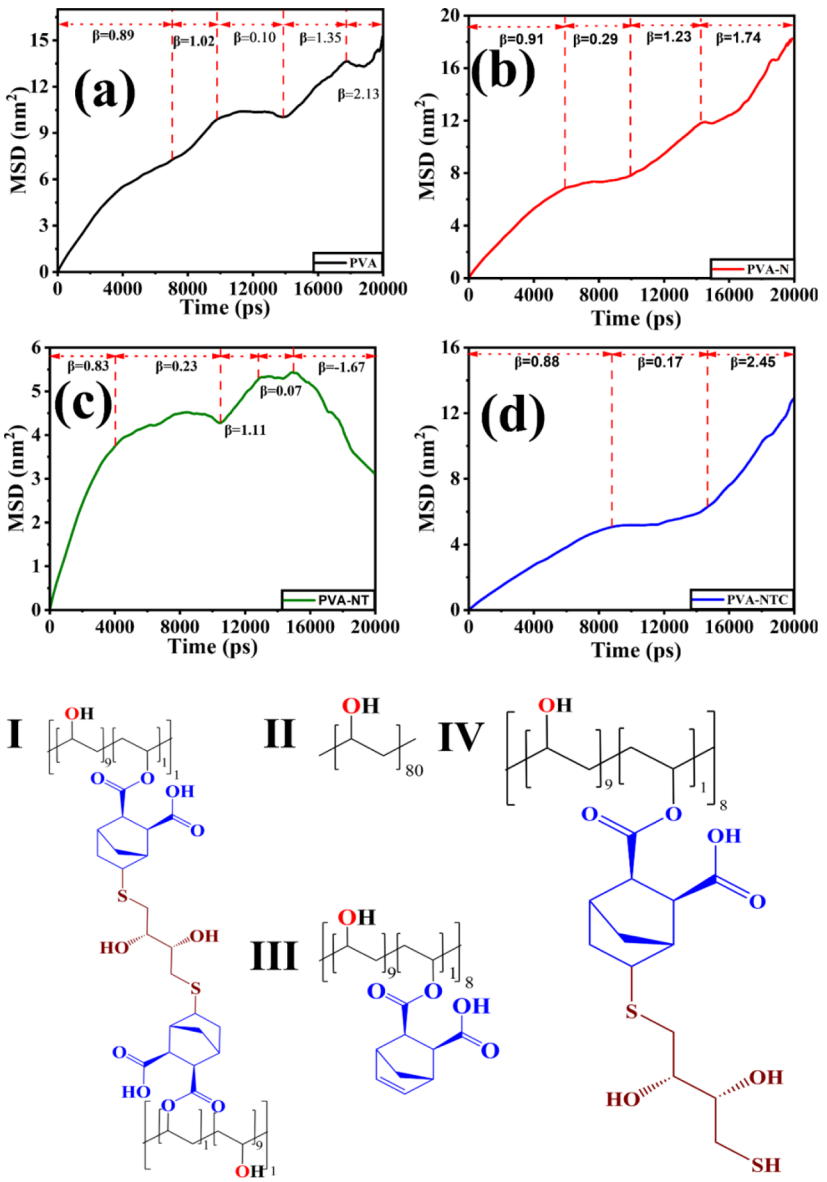


Figure 9. MSD curve of oxygen atoms in the polymer backbone. (a) MSD curve of PVA, PVA−N, and PVA−NT polymer chain; (b) MSD curve of the PVA−NTC polymer chain, (I) schematic diagram of PVA−NTC, (II) schematic diagram of PVA, (III) schematic diagram of PVA−N, and (IV) schematic of PVA−NT.

linear MSD curve with respect to time until 6 ns, but the logarithmic slope of the MSD curve is less than 1. It signifies the subdiffusive behavior of the polymer chain. This is because of the entanglement of polymer chains with each other and with the solvent molecules. After 6 ns, the MSD curve is nearly linear with respect to the time axis until 10 ns. During this period, the polymer chains were intertwined even further, creating a complex web with the solvent molecules. The entanglements persisted, continuing to impede the individual polymer chains and leading to subdiffusion. At the end of this time period, the polymer disentangles with each other. As a result, after 10 ns, the MSD curve logarithmic slope increases significantly. After 10 ns, the MSD curve becomes concave upward, which signifies the superdiffusive behavior until 14 ns. During this time period, the hydroxyl moiety disentangles with another polymer chain. As a result, the hydroxyl moieties move freely. After 14 ns, the hydroxyl moieties show a further nonlinear behavior, which is shown from the MSD curve. During this time period, the polymer chains completely disentangle with each other and solvent molecules as the movement of hydroxyl moiety is more directed. The PVA−NT (Figure 9c) polymer shows the lowest MSD value among other polymer chains. The MSD curve shows five distinct regions. Initially, the MSD curve shows the linear behavior with respect to time until 4 ns. The logarithmic MSD slope value is less than 1, which suggests the subdiffusive behavior of the polymer chain. This is because of the polymer chain entanglement. After the 4 ns, the MSD curve becomes concave downward, which signifies the subdiffusive behavior of the polymer chain until 10.3 ns. During this time period, the hydroxyl moiety of the PVA−NT chain experiences more interaction with polymers and solvents. After 10.3 ns, the PVA−NT chain hydroxyl moieties start to move until 12.8 ns. As a result, the MSD curve slope increases and shows a linear behavior. The logarithmic slope of the MSD curve is greater than 1, which signifies the superdiffusive behavior of the polymer chain. Due

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to the conformational change of the PVA−NT polymer, a superdiffusive behavior is observed. After 12.8 ns, the MSD curve is parallel to the time axis, and the MSD slope decreases until the end of the dynamics. This is because of the entanglements of the polymer structures with each other.

Figure 9d presents the MSD curve of oxygen atoms in the PVA−NTC polymer chain, which shows initially linear behavior for 8.8 ns. The linear nature signifies the random motion of hydroxyl moieties in an aqueous medium. The logarithmic MSD slope value is less than 1, which signifies the subdiffusive behavior of the polymer chain. After 8.8 ns, the chains undergo slower motion where the MSD increases gradually than the time curve. As a result, it shows a concave upward regime until the end of the dynamics. This curvature represented particle motion becoming restricted over time. This subdiffusive behavior of the polymer or the gel structure is commonly observed where the network experiences entanglement or physical trapping, which leads to a more complex and restricted motion.

Figure 10 shows the MSD curve of the conjugate double bond present in norbornene (marked in red) belonging to

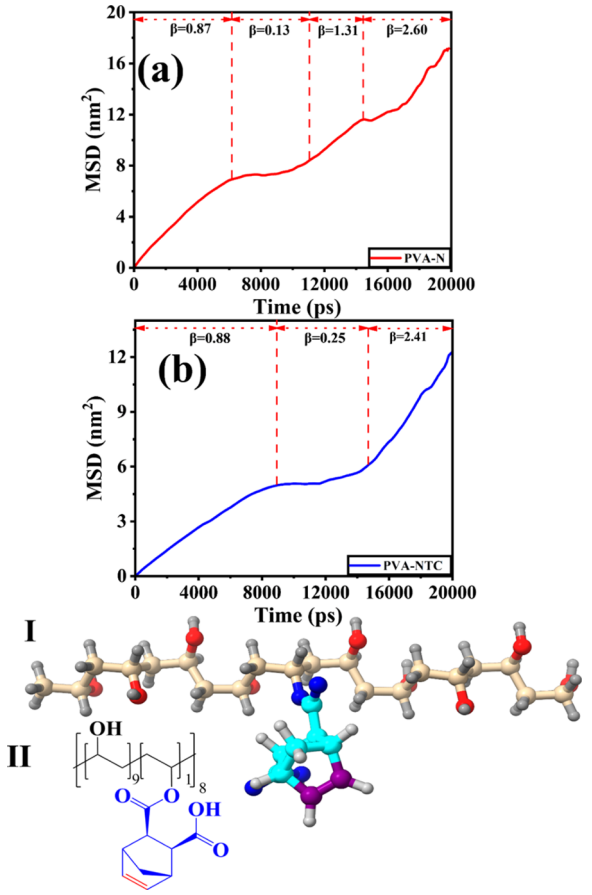


Figure 10. MSD curve of the conjugate double bond of norbornene present in (a) PVA−N and (b) PVA−NTC polymer chains. (I) 3D representation of PVA−N polymer chain and (II) schematic diagram of the PVA−N polymer chain.

PVA−N and PVA−NTC polymer chains. The PVA−N (Figure 10a) polymer chain shows the three regions in the MSD curve. The initial region is showing a linear behavior at 6 ns, which signifies the random motion of conjugate double bonds in the aqueous medium. The logarithmic MSD slope is less than 1, which signifies the subdiffusive behavior. After 6 ns, the conjugate double-bond moieties start to go slower. As a result, the MSD curve shows a nonlinear behavior. After 6 ns, the conjugate double bond shows a concave upward curve in the MSD curve up to 14.4 ns. The concave upward movement of the MSD curve indicates that the mobility of the conjugate double bonds is slower than expected for normal or linear diffusion. The conjugate double bonds experience hindered diffusion. After 14.9 ns, the concave upward regime increases until the end of the dynamics. The increasing concavity of the curve indicates that the rate of diffusion is becoming slower over time. Figure 10b displays the MSD curve of conjugate double bonds present within the PVA−NTC polymer chain. Initially, the MSD curve shows a linear behavior for a period of 9 ns. This linear behavior suggests that the conjugate double bonds are randomly moving in the aqueous medium, but the logarithmic MSD slope shows a value less than 1, which signifies the subdiffusive behavior. However, after the first 9 ns, the conjugate double bonds start to exhibit slower motion, with the MSD curve increasing slower than the time curve. Consequently, the MSD curve shows a concave upward shape until the end of the dynamics, indicating that the motion of the chains becomes progressively restricted over time.

Figure 11 shows the MSD curve of unreacted thiol moieties present in the PVA−NT polymer chain and the PVA−NTC polymer chain. Figure 11a represents the unreacted thiol moiety present in the PVA−NT polymer chain. In the initial

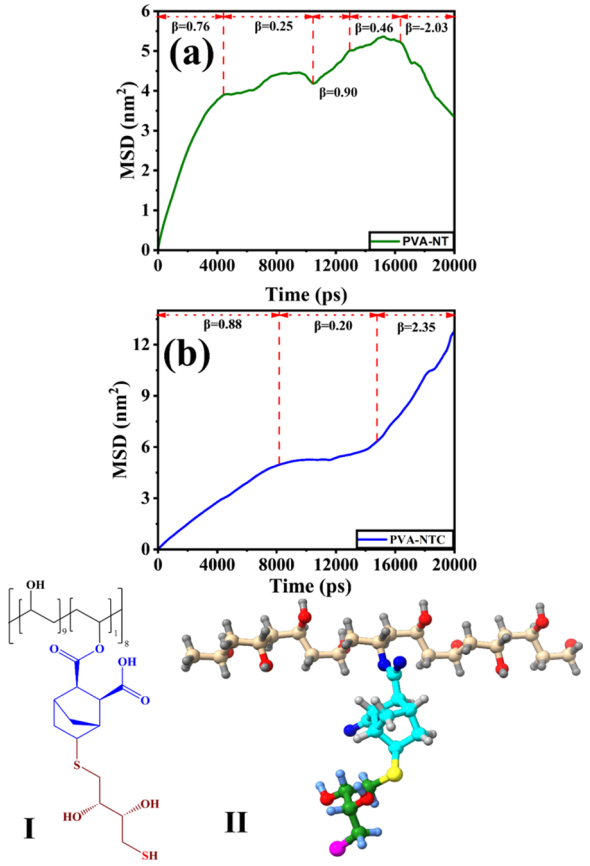


Figure 11. MSD curve of the unreacted thiol is present in (a) PVA−NT and (b) PVA−NTC polymer chains. (I) Schematic diagram of the PVA−NT polymer chain. (II) 3D representation of the PVA−NT polymer chain.

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linear regime, the MSD curve shows a linear relationship between the MSD and the time for 4.4 ns, indicating that the unreacted thiol moieties are undergoing a random motion. The logarithmic MSD slope value is less than 1, which signifies the subdiffusive behavior of the thiol moiety in the polymer chain. As time progresses, the behavior of the unreacted thiol moieties in hydrogel systems may change due to interactions with the hydrogel network. Hydrogel networks can exert a variety of physical and chemical forces on the unreacted thiol moieties such as steric hindrance, electrostatic repulsion, and chemical interactions. These forces can lead to a decrease in the mobility of unreacted thiol moieties, resulting in a concave downward regime in the MSD curve of up to 16.3 ns. In this regime, the MSD curve no longer follows a linear relationship but instead begins to curve downward, indicating that the unreacted thiol moieties no longer undergo a purely diffusive motion. After 16.3 ns, the MSD curve shows a negative slope until the end of the dynamics. This is because the thiol moiety in the PVA−NT polymer chain experiences motion in a particular direction. This directed motion can lead to a negative slope in the MSD curve. Figure 11b represents the MSD curve of the unreacted thiol moiety present in the PVA−NTC polymer chain, initially exhibiting a linear behavior for a duration of 8.1 ns. This linear behavior is indicative of the random motion of unreacted thiol moieties in the aqueous medium. The logarithmic MSD slope value is less than 1, which signifies the subdiffusive behavior. However, after 8.1 ns, the unreacted thiol moieties start undergoing slower motion, leading to a slower increase in MSD than the corresponding time curve. As a result, the MSD curve becomes concave upward until the end of the dynamics. This upward curvature signifies that the motion of the unreacted thiol moiety becomes increasingly restricted over time. Such MSD behavior is common in polymer or gel structures where the network experiences entanglement or physical trapping, resulting in more complex and restricted motion.

4. CONCLUSIONS

All-atom MD simulation is employed to investigate the behavior of a chemically cross−linked PVA hydrogel with a high hydration degree. The dynamics reveals that norbornene plays a crucial role in stabilizing the PVA polymer chains and contributes to the overall stability of a hydrogel. Analysis of RDF reveals the formation of small-sized clusters with the interactions between the PVA polymer chain and water. Further, a deeper understanding of the interactions between unreacted thiol and ene moieties in the PVA−NTC polymer chain has been obtained. The largest adhesive property of the PVA−NT hydrogel among the four cross−linked networks is confirmed by the calculation of the H−bond, while PVA−NTC showed the strongest cohesive properties among all hydrogels. The swelling ratio of the PVA−N hydrogel is higher than that of other hydrogels. The mobility of chains is further explored by the MSD curves. All the hydrogels have shown at least two diffusive regimes, namely, subdiffusive and diffusive. Overall, the results of the simulations provide a deeper understanding of the interactions between functionalized PVA hydrogels and water, shedding light on the structural and transport properties of the hydrogel. This information will be valuable in the design of hydrogels with specific properties for various applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.3c01948.

Detailed information on simulation systems, initial geometry of polymers, and dynamic behavior of hydrogels at different time intervals (PDF)

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Notes

The authors declare no competing financial interest.

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